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EDA

- Exploratory data analysis should be used to A.
help you search for patterns in your data.
B. spot serious defects in your data that may warrant taking corrective action.
C. help determine whether assumptions of the inferential tests you intend to use may have been violated.
D. **all of the above**
- A bar graph is the best graph to use when
A. your dependent variable was measured on at least a ratio scale.
B. **your independent variable is categorical.**
C. your independent and dependent variables are both continuous.
D. you want to show ordered trends in your data.
- To show a functional relationship between your independent and dependent variables, the graph of choice would be a
A. **line graph.** B. histogram. C. pie chart D. scatterplot.
- The Spearman Rank Order Correlation is used when A.
your data are scaled on an ordinal scale.
B. your data are scaled on an interval scale.
C. one measure is scaled on a nominal scale and the other on an ordinal scale.
D. one measure is scaled on a in interval scale and the other on an ordinal scale
- In which of the following situations would you not want to use a Pearson correlation coefficient?
A. **when the relationship between variables is nonlinear**
B. when both of your variables are measured on at least an interval scale
C. when the variances of your distributions are very similar D. all of the above
- A curve showing a functional relationship that starts off flat, becomes progressively steeper, and shows a single direction of change is
A. **negatively accelerated.** B. monotonic
C. positively accelerated. D. both b and c
- A _____ distribution has most scores collected about the center and is symmetrical about its midpoint.
A. functional B. normal C. monotonic D. bimodal

8. _____ are used to represent category values (e.g., gender) as values.
 A. Unstacked formats B. Dummy codes C. Stacked formats D. Codes
9. A functional graph that shows a uniformly increasing or decreasing functional relationship is said to be **A. monotonic.** B. negatively skewed. C. normal. D. positively skewed.
10. If you have discrete group data, such as months of the year, age group, shoe sizes, and animals. Which is best to explain?
 A. Boxplot B. histogram C. bar D. scatterplot
11. Which graph is better used when data needs to be classified or categorized?
A. stack bar B. Pie chart C. histogram D. None of the above
12. Which is best to explain a relationship between target and feature?
A. scatterplot B. bar C. Pareto chart D. all of the above
13. How can you check for outliers in data set?
 Using scatterplot B. Using histogram C. Using Boxplot **D. all of the above**
14. From which plot you will come to the distribution of the target variable?
A. histogram B. pie chart C. bar D. Pareto chart
15. True/False: The quantile-quantile (qq) plot is a graphical technique for determining if two data sets come from populations with a common distribution.
A. True B. False
16. True/False: In Boxplot the middle line inside the box displays the mean of the distribution. **A. True** **B. False**
17. True/False: For Numeric vs Numeric data scatterplot is the best representation. **A. True** B. False
18. True/False: For Bivariate data, correlogram or corr plot show the correlation of each variable. **A. True** B. False
19. True/False: the height of the bar corresponds to the value of each category. **A. True** B. False
20. True/False: The height of the resulting Stacked Bar shows the combined result of the groups. **A. True** B. False

Numpy, Pandas and Data Visualization

- 1) Pandas is designed to work with _____ data.
 A. Relational B. Labeled **C. Both of these** D. None of these
- 2) DataFrame is a _____ labeled data structure.
 A. 1dimensional **B. 2dimensional** C. 3dimensional D. ndimensional

- 3) Pandas does easy handling of missing data in floating point as well as nonfloating point data? **A. True** B. False
- 4) Columns can be deleted and inserted from:
A. DataFrame B. Higher dimensional objects.
C. All of the above D. None of the above
- 5) Shape property in pandas is used to
A. Visualise the distribution of the data
B. See the number of rows and columns of the data
C. Visualise the shape of skewness of the data
D. See the spread of data (mean, median etc.)
- 6) The _____ method allows us to retrieve rows and columns by position.
A. head B. getloc **C. iloc** D. locate
- 7) Pivot table can aggregate the data and summarize it by grouping the columns **A. True** B. False
- 8) _____ is a convenient method for combining the columns of two potentially differently indexed DataFrames into a single result DataFrame.
A. Concatenate B. Merge **C. Join** D. Collaborate
- 9) Dimensions should match along the axis you are _____ on.
A. concatenating B. merging C. joining D. collaborating
- 10) Series can have axis labels and it can be indexed by a label
A. True B. False
- 11) Matplotlib is a _____ library for data visualisation.
A. 1dimensional **B. 2dimensional** C. 3dimensional D. ndimensional
- 12) Select the proper sequence to create a plot:
A. Set plot parameters, import required libraries, define the required dataset, display plot.
B. Define the required dataset, set plot parameters, import required libraries, display plot. C.
Set plot parameters, define the required dataset, import required libraries, display plot.
D. Import required libraries, define the required dataset, set plot parameters, display plot.
- 13) The plt.subplots() object acts as a more automatic axis manager?
A. True B. False
- 14) To avoid the overlapping of subplots we use
A. fig.tight_layout() B. sub.tight_layout() C. plt.tight_layout()
- 15) We cannot create a horizontal bar plot in matplotlib
A. True **B. False**

16) We use `plot.barh()` to adjust the height of the plot

- A. True **B. False**

Explanation: We use it to create a horizontal barplot

17) We use _____ to create a horizontal bar plot.

- `axesh.bar()` B. `haxis.bar()` **C. `axes.barh()`** D. `hor.barh()`

18) _____ is a visualisation library that provides a highlevel interface to draw attractive statistical graphics.

- A. Scrapy **B. Seaborn** C. Airborn D. Statistica

Matplot and seaborn

1. The plot method on Series and DataFrame is just a simple wrapper around :

- A. `gplt.plot()` **B. `plt.plot()`** C. `plt.plotgraph()` D. none of the Mentioned

Explanation: If the index consists of dates, it calls `gcf().autofmt_xdate()` to try to format the xaxis nicely.

2. Point out the correct combination with regards to kind keyword for graph plotting:

- A. 'hist' for histogram B. 'box' for boxplot
C. 'area' for area plots D. all of the Mentioned

Explanation: The kind keyword argument of `plot()` accepts a handful of values for plots other than the default Line plot.

3. Which of the following value is provided by kind keyword for barplot ?

- A. `barh`** B. `kde` C. `hexbin` D. none of the Mentioned

Explanation: bar can also be used for barplot.

4. You can create a scatter plot matrix using the _____ method in `pandas.tools.plotting`.

- A. `sca_matrix` **B. `scatter_matrix`** C. `DataFrame.plot` D. all of the Mentioned

Explanation: You can create density plots using the `Series/DataFrame.plot`.

5. Point out the wrong combination with regards to kind keyword for graph

- plotting: A. 'scatter' for scatter plots **B. 'kde' for hexagonal bin plots** C.
'pie' for pie plots D. none of the Mentioned

Explanation: kde is used for density plots.

6. Which of the following plots are used to check if a data set or time series is random ?

- A. Lag** B. Random C. Lead D. None of the Mentioned

Explanation: Random data should not exhibit any structure in the lag plot.

7. Plots may also be adorned with error bars or tables.

- A. True** B. False

Explanation: There are several plotting functions in `pandas.tools.plotting`.

8. Which of the following plots are often used for checking randomness in time series ?

- A. Autocausation B. Autorank **C. Autocorrelation** D. None of the Mentioned

Explanation: If time series is random, such autocorrelations should be near zero for any and all timelag separations.

9. _____ plots are used to visually assess the uncertainty of a statistic.

- A. Lag B. RadViz **C. Bootstrap** D. None of the Mentioned

Explanation: Resulting plots and histograms are what constitutes the bootstrap plot.

10. Andrews curves allow one to plot multivariate data.

- A. True** B. False

Explanation: Curves belonging to samples of the same class will usually be closer together and form larger structures.

